Pandita's Aether-DCTWIN

AI Models, System Architecture & Technology Stack Overview

1. System Architecture (100% Facility-Side Boundary)

LAYER 1 • FIELD OT
Facility Meters
Read-only BACnet/IP, Modbus TCP, SNMPv3, Sparkplug B & CSV exports.

→

LAYER 2 • STREAM
FastAPI & Historian
1.5s WebSocket push, YAML hall blueprints & DuckDB + Parquet 24h store.

→

LAYER 3 • INTELLIGENCE
5-Layer Hybrid AI
RLS calibration, IsolationForest, SciPy MPC, GraphRAG & LLM tools.

→

LAYER 4 • OPERATOR
3D Twin & Control
React + Three.js hall, 10k-GPU planner, Copilot & safe BMS fallback.

Core AI Safety Principle — Why LLMs Never Touch Cooling Hardware Directly: In critical data halls, black-box language models should never guess temperatures or write raw setpoints. In Aether-DCTWIN, LLMs explain insights and call read-only inspection tools, while **deterministic thermodynamics and constrained mathematical solvers** verify every setpoint against ASHRAE limits (zero agents on tenant servers).

2. What AI We Use — Five Purpose-Built Intelligence Layers

1. Multi-Model Agentic Copilot (Function-Calling LLMs)

Claude Sonnet • Gemini 2.5 Pro • GPT-4o • Ollama

Supports frontier cloud models via `OpenRouter` / native APIs or air-gapped on-prem `Llama 3.3 70B` & `DeepSeek-R1` via `Ollama`. Rather than guessing, the LLM runs structured JSON tool calls across **7 deterministic inspection, DuckDB SQL, What-If, and GraphRAG tools** (with an offline fallback) so every number is auditable.

2. Topology-Aware GraphRAG (Alarm Correlation & Automated MOPs)

NetworkX Knowledge Graph + SOP Engine

Models the live physical dependency graph (`UPS → Transformer → PDU → Rack ← CDU / CRAC`). Traverses upstream/downstream links to collapse 50+ cascading rack alerts into **1 root-cause equipment fault**, calculate N-1 blast radius, and draft step-by-step ASHRAE/LOTO maintenance checklists (`MOPs`).

3. Physics-Guided Anomaly Detection & Thermal Forecasting

scikit-learn IsolationForest + State-Space

Pairs unsupervised multivariate `IsolationForest` outlier detection (rack inlet °C, air/liquid ΔT, thermal ratio, load ratio) with thermodynamic energy-balance residuals and **5 / 15 / 30-minute state-space trajectory forecasts**—separating real blower trips or containment leaks from normal AI workload spikes.

4. Self-Calibrating Digital Twin (Online Parameter Learning)

NumPy Recursive Least Squares (RLS / Kalman)

Online covariance filter continuously learns each zone's effective thermal resistance (`R_th`) and thermal capacitance (`C_th`) from standard PDU and cooling meters—detecting gradual `CDU` strainer clogging and heat-exchanger fouling days before static threshold alarms trip.

5. Constrained Autonomous Cooling & Battery Control (Auto-Pilot)

SciPy SLSQP MPC + Control Barrier Functions

Solves 30-minute receding-horizon cooling and UPS battery peak-shaving setpoints using Sequential Least Squares Programming (`SLSQP`) inside hard **Control Barrier Functions** (`ASHRAE ≤ 27°C` cold inlet, dew-point condensation margin, and `≤ 4%/min` VFD fan/pump slew limits).

3. Deterministic Physics & 10,000-GPU Capacity Planning Engine

THERMODYNAMICS & POWER

First-Principles Physics SDK

Unit-tested Python SDK (`backend/domain/`) computing Magnus-Tetens dew point, chiller COP & wet-bulb free-cooling curves, cubic VFD affinity laws (`P ∝ RPM³`), and 120° 3-phase AC neutral current phasors.

MULTI-HALL AI SCALE-OUT

10k-GPU Cluster Planner

Sizes 2k–25k GPU deployments (`GB200 NVL72` , `HGX H200`): calculates halls needed, 87/13 liquid-to-air heat split (`21,400 LPM`), 415V busways, slab load (`1,630 kg/rack`), and <100 ms BESS step-load buffers.

COUNTERFACTUAL SIMULATION

Offline "What-If" Sandbox

Clones live hall state in memory to simulate +160 kW liquid rack additions, air-to-liquid row retrofits, summer heatwaves, or N-1 cooling unit trips in <50 ms before touching floor hardware.

4. Full Application, Data & Protocol Technology Stack

LAYER	CORE TECHNOLOGIES & HOW THEY ARE USED
3D Visualization & UI	`React 18` , `TypeScript` , `Three.js` (`@react-three/fiber` , `@react-three/drei`), `Tailwind CSS` , and `Recharts` — hardware-accelerated 3D hall, 42U rack inspector, Dark/Light themes, and live trend charts.
Backend & Streaming API	`Python 3.11+` , `FastAPI` , `Uvicorn` , `Pydantic v2` , and `PyYAML` — async REST endpoints, 1.5-second WebSocket state push, and declarative YAML site blueprints (zero hardcoded asset IDs).
Time-Series Historian	Embedded `DuckDB` + `Apache Parquet` (`pyarrow`) — columnar 24-hour telemetry store powering the UI's DVR time-travel replay scrubber and sub-10 ms SQL analytics with zero external DB setup.
AI, ML & Scientific Computing	`NumPy` (RLS calibration), `SciPy` (SLSQP constrained MPC), `scikit-learn` (IsolationForest), `NetworkX` (GraphRAG topology), and multi-provider `OpenRouter / OpenAI / Anthropic / Gemini / Ollama` tool-calling loop.
Field OT Connectors	Pluggable adapter interface supporting `BACnet/IP` (Priority 8 supervisory writes with atomic `NULL` relinquish to Priority 16 local BMS), `Modbus TCP` (CDUs), `SNMPv3` (PDUs), `Sparkplug B` , and `BYOD CSV` .
Modularity, Testing & Packaging	Auto-discovered backend/UI feature plugins, `Pytest` physics & security boundary test suite, `GitHub Actions` CI, single-artifact container (`Dockerfile` + `docker-compose.yml`).

5. Flexible Deployment Modes (Cloud, Hybrid, or Air-Gapped On-Prem)

MODE 1 • OFFLINE AUDIT

BYOD CSV / Historian Export

Upload 30 days of BMS & PDU CSV logs to quantify cooling waste, fouling drift, and stranded kW offline in hours.

MODE 2 • READ-ONLY SHADOW

On-Prem + Local Ollama LLM

Deploy inside the facility VLAN reading live meters with an on-prem LLM—zero external cloud egress and zero floor changes.

MODE 3 • CLOSED-LOOP

Supervised Auto-Pilot

Enable bounded BACnet setpoint tuning with continuous thermal runway tracking and instant 1-click fallback to local BMS defaults.